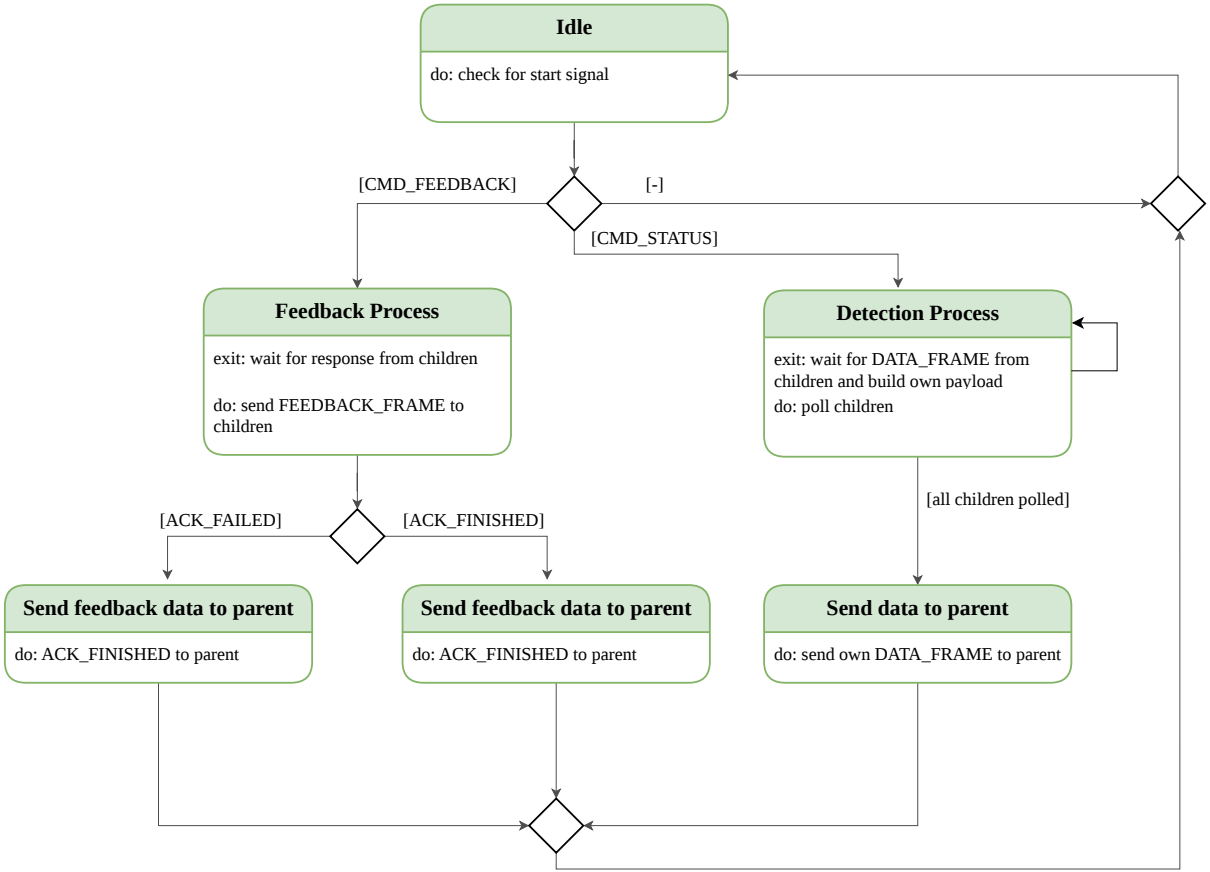
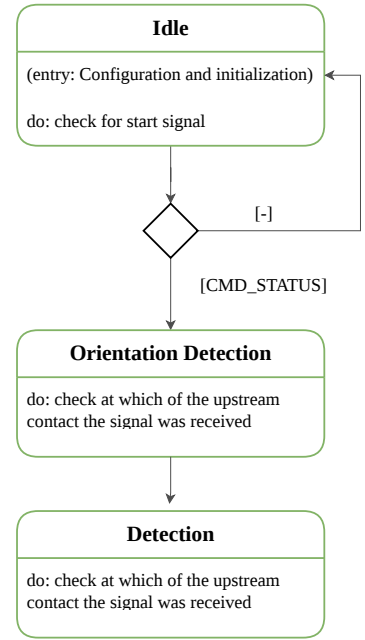
a)


Idle
do: check for start signal
[CMD_FEEDBACK]
[-]
[CMD_STATUS]
Feedback Process
exit: wait for response from children
do: send FEEDBACK_FRAME to children
Detection Process
exit: wait for DATA_FRAME from children and build own payload
do: poll children
[ACK_FAILED]
[ACK_FINISHED]
[all children polled]
Send feedback data to parent
do: ACK_FINISHED to parent
Send feedback data to parent
do: ACK_FINISHED to parent
Send data to parent
do: send own DATA_FRAME to parent

b)


Idle
(entry: Configuration and initialization)
do: check for start signal
[-]
[CMD_STATUS]
Orientation Detection
do: check at which of the upstream contact the signal was received
Detection
do: check at which of the upstream contact the signal was received